

Future Plans for the CMS Workload Management System

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on behalf of the CMS Workload Management Team

Conditions Preparation Workshop 2026

Outline

- Goals
- Current status
- Future plans
- Case of study: Condition Workflows
- Final remarks

These slides include diagrams from previous presentations.
Credit goes especially to the WM and GlideinWMS teams.

Outline

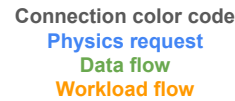
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Goals for this talk

- Share latest developments and plans for the CMS Workload Management System
- Bring awareness of the (future) capabilities
- Attempt to increase synergy towards common solutions

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WMS: Current Status

There are at least two Workload Management Systems (WMS) in CMS:

- CRAB: handling distributed analysis (and private MC)
- WMCore: handling data reconstruction and official MC simulation

Additionally, there is a WMCore “flavor” specifically for Tier-0 data processing (Repack, Express, PR, Calibration).

- We will probably hear more on this today with Antonio's [talk](#).

WMS: Current Status - Workflow Topology

Workflows are represented as Directed Acyclic Graph (**DAGs**), where:

- **edges** represent data dependency;
- **nodes** represent processing.

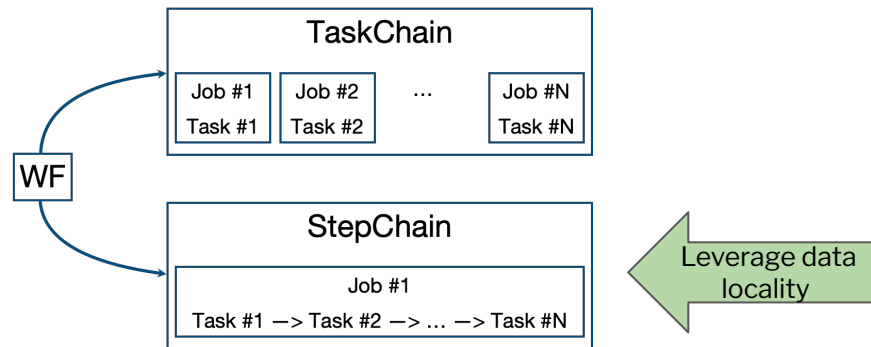
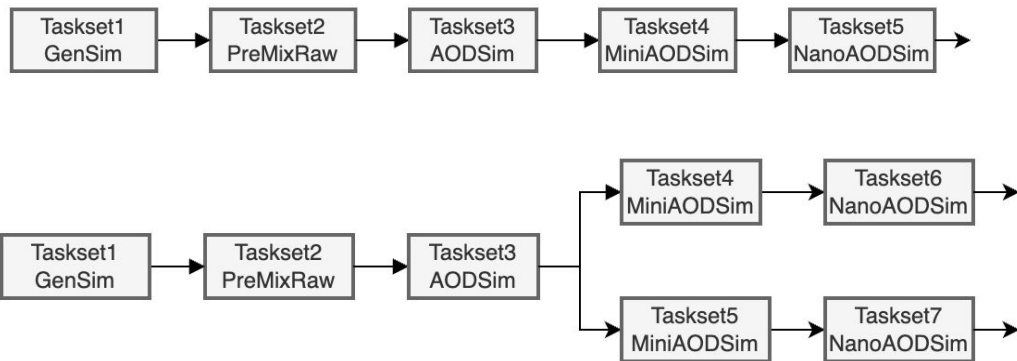
Currently supporting **linear** and **fork** topologies; **join** in the future.

Each taskset materializes between 1 and millions of grid jobs; usually with a target wallclock time.

TaskChain: completely independent.

StepChain: completely grouped.

Hybrid: any variation of the two above.



WMS: Current Status

In 2024-2025, WMCore went through a review, and the [outcome](#) was that it is not a sustainable solution for HL-LHC.

- Leading to a **freeze of the project** (as of Sept/2025), as the team engage in the future CMS WMS
- Community solutions have been considered, with emphasis on:
 - PanDA (used by Atlas and others), Dirac(X) (used by LHCb and others) and an evolution of WMCore (used solely by CMS).
 - After months of discussions and investigations, [CMS decided](#) to proceed with the DiracX solution.

The future CMS WMS (DiracX-based) has a broader scope, overarching:

- Tier-0 processing, data reconstruction, MC simulation and distributed user analysis.

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WMS: Future Plans and Core Requirements

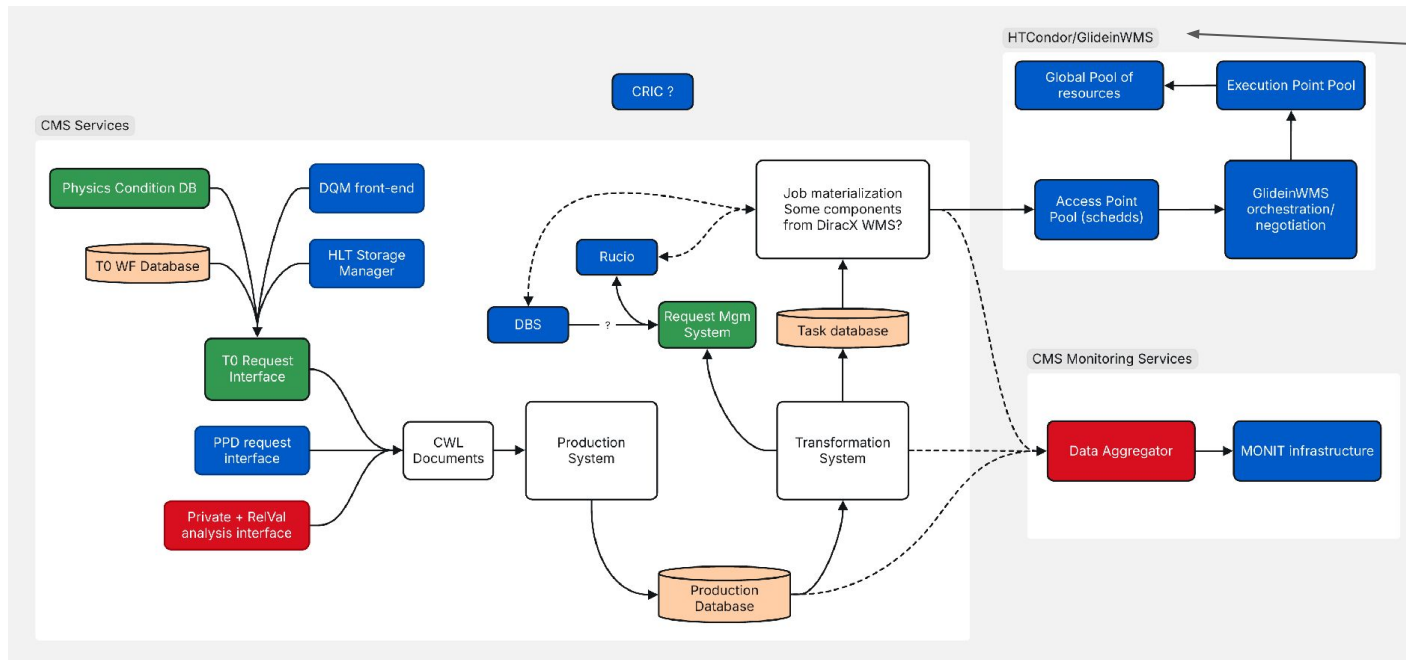
After meeting with many groups across CMS organizations (we are thankful for this), we have identified the main [requirements](#) of the future CMS WMS.

Pillars	CMS needs
Workflow lifecycle & policy	Declarative requests; flexible workflow graphs; priorities/fair-share policies; ownership boundary; automated validation recovery, closeout, and cleanup
Data-awareness	Coordination with DM on data locality, placement, integrity, partial availability, and sample lifetime
T0-ready execution	Support T0 online handshake, streamer inputs, lumi handling, calibrations
Adaptive resource model	Fault-tolerant services; heterogeneous CPU/GPU/high-memory resources; scouting-based resource estimates; constraints-vs-preferences separation
Observability + reproducibility	End-to-end workflow timelines; execution/failure traceability; job queryability; persistent task configuration metadata; data provenance/parentage
Private analysis support	Analysis pre/post-processing tasks; async stageout; configurable data publication; Python/Jupyter APIs; central MC production toolset share

WMS: Future Plans and Evolving Architecture



Initial HL-LHC WMS architecture - however, pending critical decision/knowledge to be shared from the core DiracX team (ADR: Architectural Design Record).



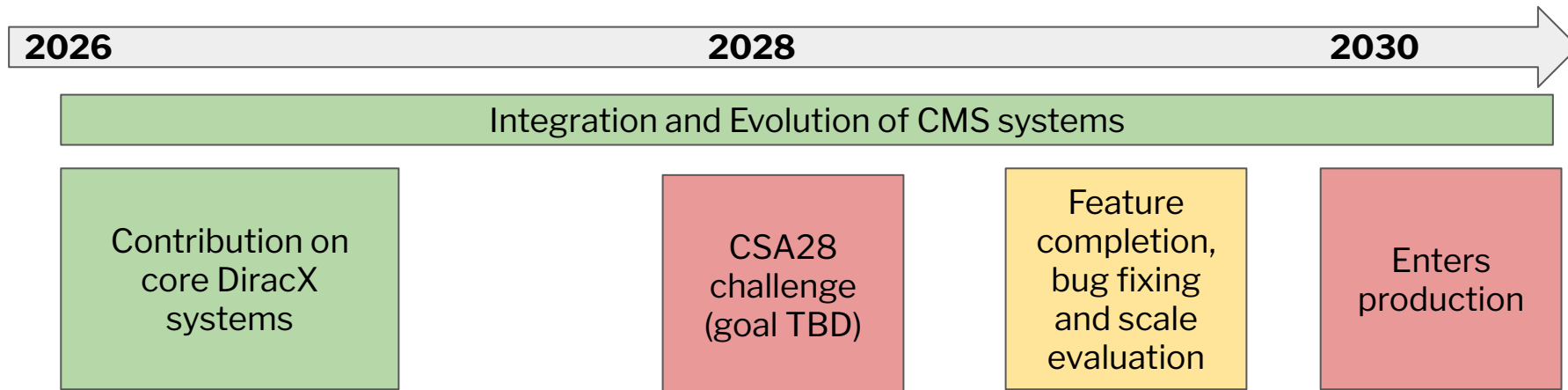
HTCondor and GlideinWMS are a core component of the current and future WMS.

Managing resources in the global and CERN pool (including HLT).

WMS: Future Plans and Evolving Architecture



Timeline under development, however, it roughly involves:



First draft also presented last week:

<https://indico.cern.ch/event/1723992/contributions/7251868/attachments/3336161/5977805/WM%20Timeline.pdf>

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Case of Study: Condition Workflows

Content based on material and previous discussion with - Jan Krzysztof C, Tomas Kello, Antonio Linares, Alejandro Salgado Manrique.

We focused on 3 family of workflows (from Run-3):

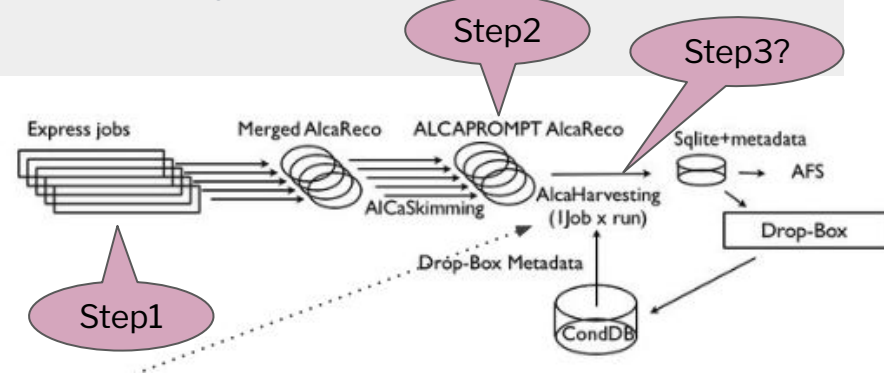
1. Prompt Calibration Loop (**PCL**)
2. Multi-Run Harvester (**MRH**)
3. Offline Automation Framework (**Auto-F**)

Case of Study: Condition Workflows

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❑ Prompt Calibration Loop (PCL)

- ❑ Running on **Express streams**
- ❑ **Step 1:** ALCARECO -> AlCaSkimming -> ALCAPROMPT
- ❑ **Step 2:** ALCAPROMPT -> Harvester -> Conditions
- ❑ **Orchestration:** Tier0Feeder daemon -> WMCore agent via WMBS Oracle DB



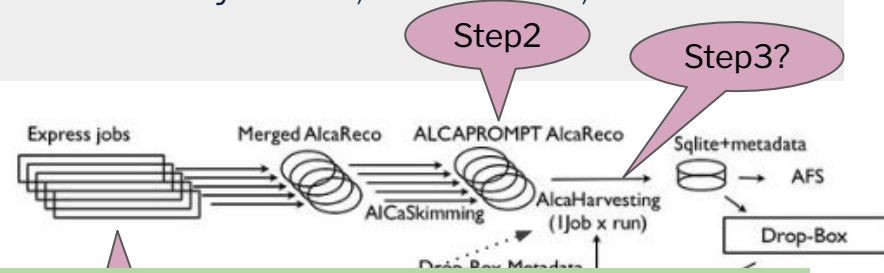
- ➔ **Multi-step** sequential workflow (step2 process unmerged files)
- ➔ **Input:** streamer files (LS granularity) - registered in StorageManager DB
- ➔ **Output:** skimmed ALCARECO ROOT files (maybe FEVT too) - registered in DBS/Rucio
- ◆ Or ALCAPROMPT ROOT files

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NOTE: as input data will be streamer files, we understand a given run/lumi might be scattered across multiple files. Thus, lumi must be completely available and all files assigned to the same grid job.

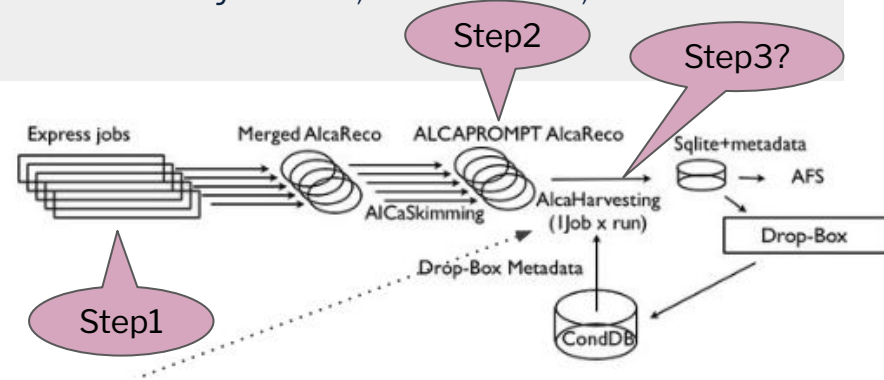
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➔ What about **Step3** (AlcaHarvesting)?

- ◆ **Input?** Can only run on data that was produced within the same workflow - unless it is managed/executed in a separated workflow
- ◆ **Output:** SQLite conditions (.db files) and DQM ROOT files uploaded to DQMGui.

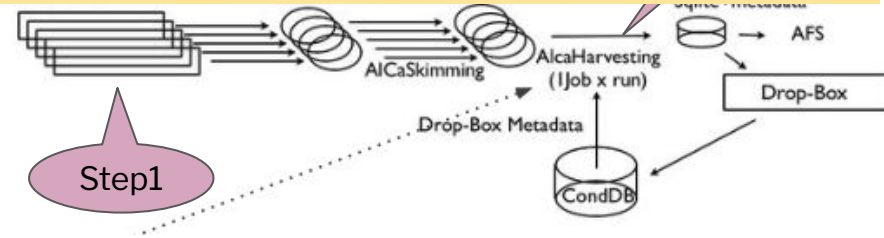
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Q1.: Will we still need periodic AlcaHarvesting? Or can it be triggered only upon run completion?

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Q2.: What are the time/latency constraints for these 3 steps?

A.: 18h for ALCARECO output; 48h for ALCAPROMPT + CondDB upload (TBC)

via **WMBS Oracle DB**

Step1

CondDB

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Q4.: It looks like another workflow (or component/daemon) is required for fetching the conditions available in the shared storage (/store/unmerged/tier0_harvest) and uploading it to CondDB. What is the input granularity (single file, all files matching a run, etc)?

Case of Study: Condition Workflows

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- ❑ **Multi-run harvester (MRH)**
 - ❑ Runs on **ALCAPROMPT** (ALCARECO possible)
 - ❑ Runs the **same harvesting workflows as PCL**
 - ❑ **Conditional harvesting** (e.g. check for enough statistics)
 - ❑ Orchestration: **RR API + T0-API + DBS API** (triggering)
Jenkins for Harvester part

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- Jenkins** for Harvester part

Q5.: Is it the same as the **AlcaHarvesting** workflow from PCL?

Q6.: Does it have the same requirements?

Q7.: How does the system knows which runs to assign to a job? Does each workflow create a single job (harvesting multiple runs)?

Case of Study: Condition Workflows

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❑ Offline Automation framework (**Auto-F**)

- ❑ Full **flexibility with input** datasets (**Express, Prompt, ReReco, MC, ...**)
- ❑ Full flexibility with workflow definition (complicated dependencies, all submission mechanisms allowed, ...)
- ❑ Full **flexibility with output** (DB conditions, monitoring plots, NTuples ...)
- ❑ Orchestration: **InfluxDB** (soon migrated to **TimeScaleDB**) + **Jenkins** (**new solution in progress**)

Need to understand:

- What is the input and which granularity
- What is the output
- Whether there any external services/constraints dependency
- What the latency is?

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Q8.: And does it have to execute in the Tier-0 scope? In other words, what are the latency requirements?

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- What is the output
- Whether there any external services/constraints dependency
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Final Remarks

Consolidation of the CMS WMS depends on DiracX core architectural design, to be finalized in the coming month.

Our goal is to provide a unified system that can handle Tier-0, Monte Carlo simulation, data reprocessing and user analysis workflows.

Currently supported Tier-0 workflows **must** also be supported in the future CMS WMS.

- Unless they are no longer relevant for HL-LHC;
- Or their requirements change (communication is key).

Questions & Discussion

CMS email list: cms-wmcore-team@cern.ch

DMWM Mattermost channel:

<https://mattermost.web.cern.ch/cms-o-and-c/channels/dmwm>

WM Dev Mattermost channel:

<https://mattermost.web.cern.ch/cms-o-and-c/channels/wm-dev>

Thank you!

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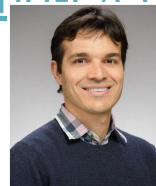
...and many others!



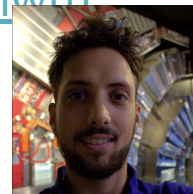
Todor Trendafilov
Ivanov



Valentin Y
Kuznetsov



Alan Malta
Rodrigues



Andrea
Piccinelli



Diego
Ciangottini



Kevin
Lannon



Marco
Mascheroni



Eric Wayne
Vaandering



???



Vijay
Chakravarty



Stefano
Belforte



???